

Towhid Ahmed

Generative AI Engineer · Open to remote — US/EU time zones

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Professional Summary

Generative AI Engineer with 1+ year of freelance experience building **production-grade LLM applications, multi-agent systems, and RAG** solutions for SaaS and e-commerce platforms. Developed 5 AI applications using **LangChain, LangGraph, and FastAPI**, achieving **35% hallucination reduction, 25% retrieval relevance improvement, and 98% test coverage**. Experienced in **RAG pipelines, LLMops, Docker, and LangSmith**.

Technical Skills

Generative AI & Agentic Systems: LangChain, LangGraph, Multi-Agent Systems, Tool Calling, Prompt Engineering, Advanced RAG, Multimodal RAG, RAG Pipeline, Retrieval Strategies, Cross-Encoder Re-ranking, Vector Databases (Pinecone, Weaviate, ChromaDB)

LLMOps & Deployment: LangSmith, LLM Guardrails, FastAPI, Docker, Streamlit, CI/CD (GitHub Actions), Render

LLMs & AI Frameworks: Llama, OpenAI API, Groq, Hugging Face Transformers, Sentence Transformers

Machine Learning & Deep Learning: Scikit-learn, Feature Engineering, Supervised & Unsupervised Learning

Programming & Tools: Python, PostgreSQL, SQL, Git, GitHub, Pandas, NumPy, Java

Software Engineering: Data Structures & Algorithms (Solved 400+ problems), OOP, LLM System Design

Professional Experience

Freelance Generative AI Engineer | Self-Employed | Jun 2025 – Present

AI Document Assistant | June 2026 – July 2026

- Developed a **3-agent LangGraph** workflow with Pinecone semantic search and citation-based RAG, **reducing hallucinations**.
- Engineered **7 REST APIs** using FastAPI, LangChain, OpenAI Embeddings, PostgreSQL, and Streamlit.
- Built a production-ready system processing **1,000+ documents** with structured logging, error handling, and session management.

AI Sales & Customer Support Assistant | Aug 2025 – Sep 2025

- Architected a **multi-agent LangGraph** pipeline (Intent, Retrieval, Response) with Pinecone RAG for automated customer interactions.
- Developed a FastAPI backend with **6+ REST APIs**, a Streamlit interface, knowledge base upload, and conversational session memory.
- Integrated Groq LLM, local Hugging Face embeddings, and Pinecone Serverless to **optimize inference** cost for the AI assistant.

Selected Projects

Multi-Agent AI System — Python, LangGraph, LangChain, FastAPI, Pinecone, LangSmith, Streamlit, Docker, Render, HuggingFace, Tavily GitHub | Demo

- Designed **4-agent LangGraph** workflow routing queries between **RAG** and **web search** based on intent and complexity.
- Built **FastAPI** backend with async execution, **7 core APIs** (/chat, /upload, /health, /metrics) optimized for **Render**.
- Integrated **Tavily with DuckDuckGo fallback** and **LangSmith tracing** for reliable web retrieval and **observability**.

Multimodal RAG — Python, FastAPI, Streamlit, Pinecone, Groq, Llama 3.3, Faster-Whisper, RAG, Vector Search, LangSmith, Docker, Git, GitHub, REST API, Async Programming, Caching GitHub

- Designed production multimodal retrieval (PDF/audio/video) **improving relevance by 25% and reducing hallucinations by 35%**.
- Implemented two-stage retrieval (**semantic chunking + cross-encoder re-ranking**) with **<2s latency** via **caching**.
- Tech: **Groq Llama 3.3, BAAI embeddings, Pinecone, FastAPI, Streamlit, Docker**; 76-unit test coverage.

Clinical RAG Assistant — Python, FastAPI, LangGraph, RAG, Pinecone, LLM Guardrails, Tavily, Streamlit, LangSmith, Docker, JWT GitHub

- Created **clinical AI** using **4-agent LangGraph** pipeline (Router+Safety+Retrieval+Answer) with **RAG and web search**.
- Implemented **multi-layer safety system** (input filters, retrieval validation, output sanitization) with JWT auth.
- Achieved **98/98** passing tests across **guardrails, RAG evaluation, and API workflows**.

Education and Certificates

Education: Computer Science Engineering(CSE), Chittagong University of Engineering & Technology, Bangladesh

Certifications: • Generative AI & Agentic AI — Krish Naik (Hands-on LLM systems & agents)

• LLMops — Udemy (Deployment, monitoring, CI/CD for LLM apps) • Data Structures & Algorithms — TakeUForward